

RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Page and volume buttons



File: user_controls.kicad_sch

ESP32-C6 radio



File: radio_esp32.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824–#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

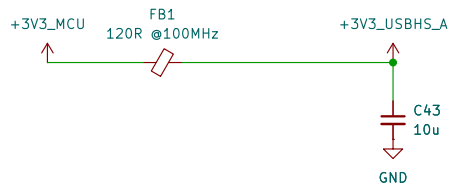
Footprint and PCB qualification are deferred by scope.

Power button and source control

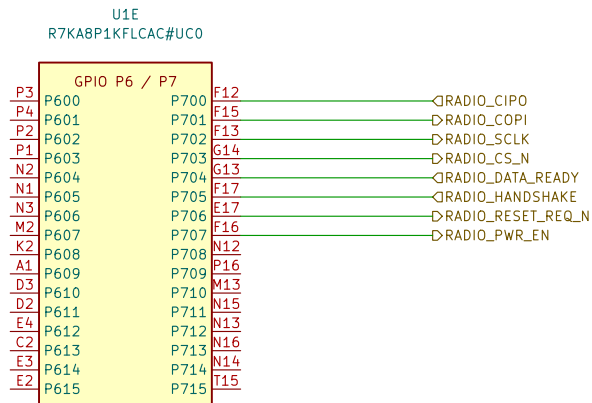
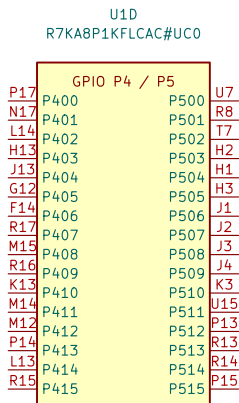
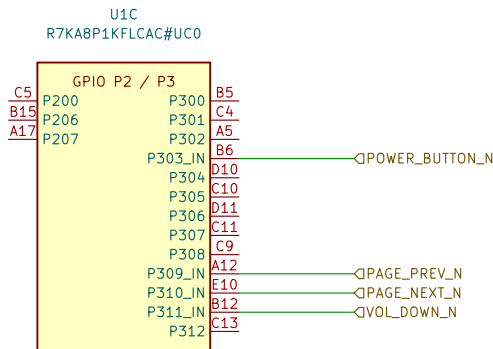
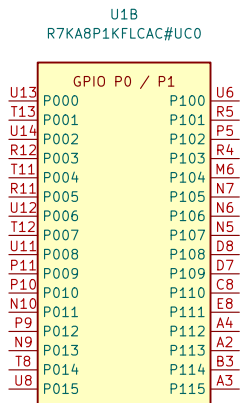


File: power_button.kicad_sch

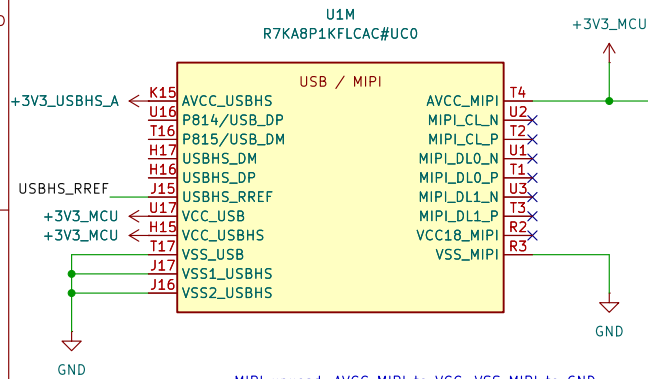
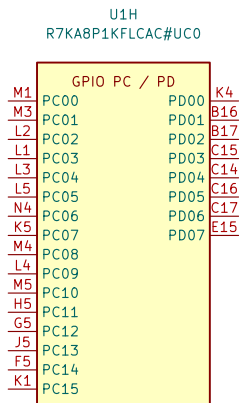
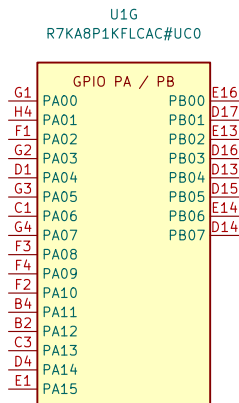
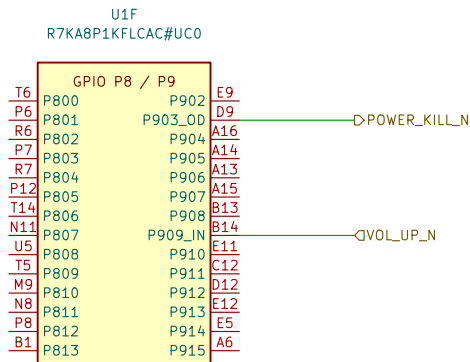
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



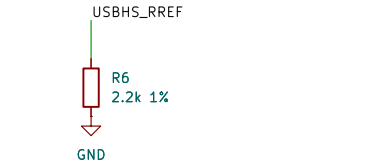
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



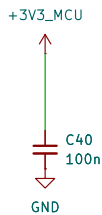
RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



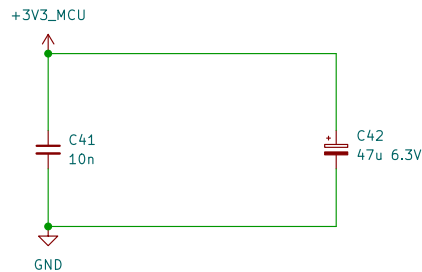
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

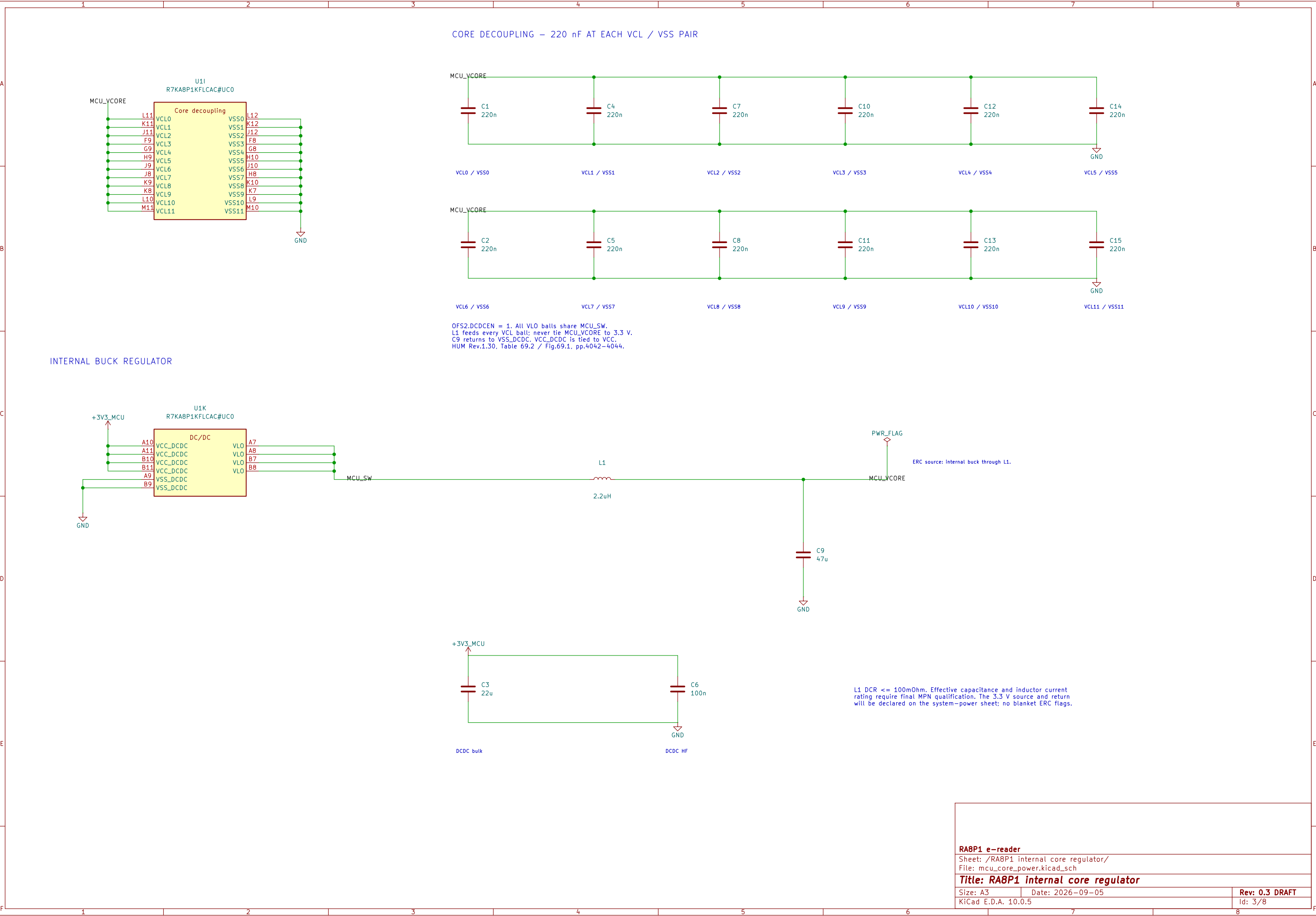
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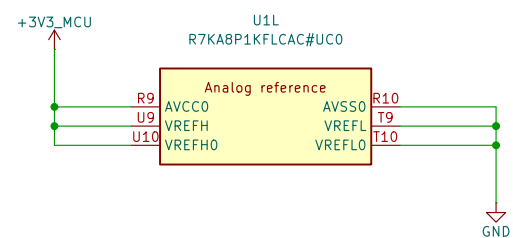
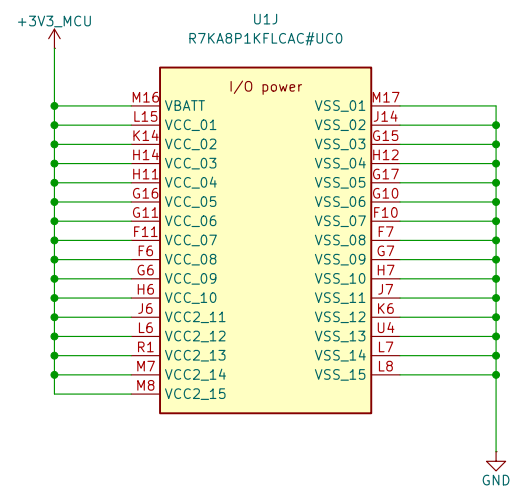
Date: 2026-09-05

Rev: 0.3 DRAFT

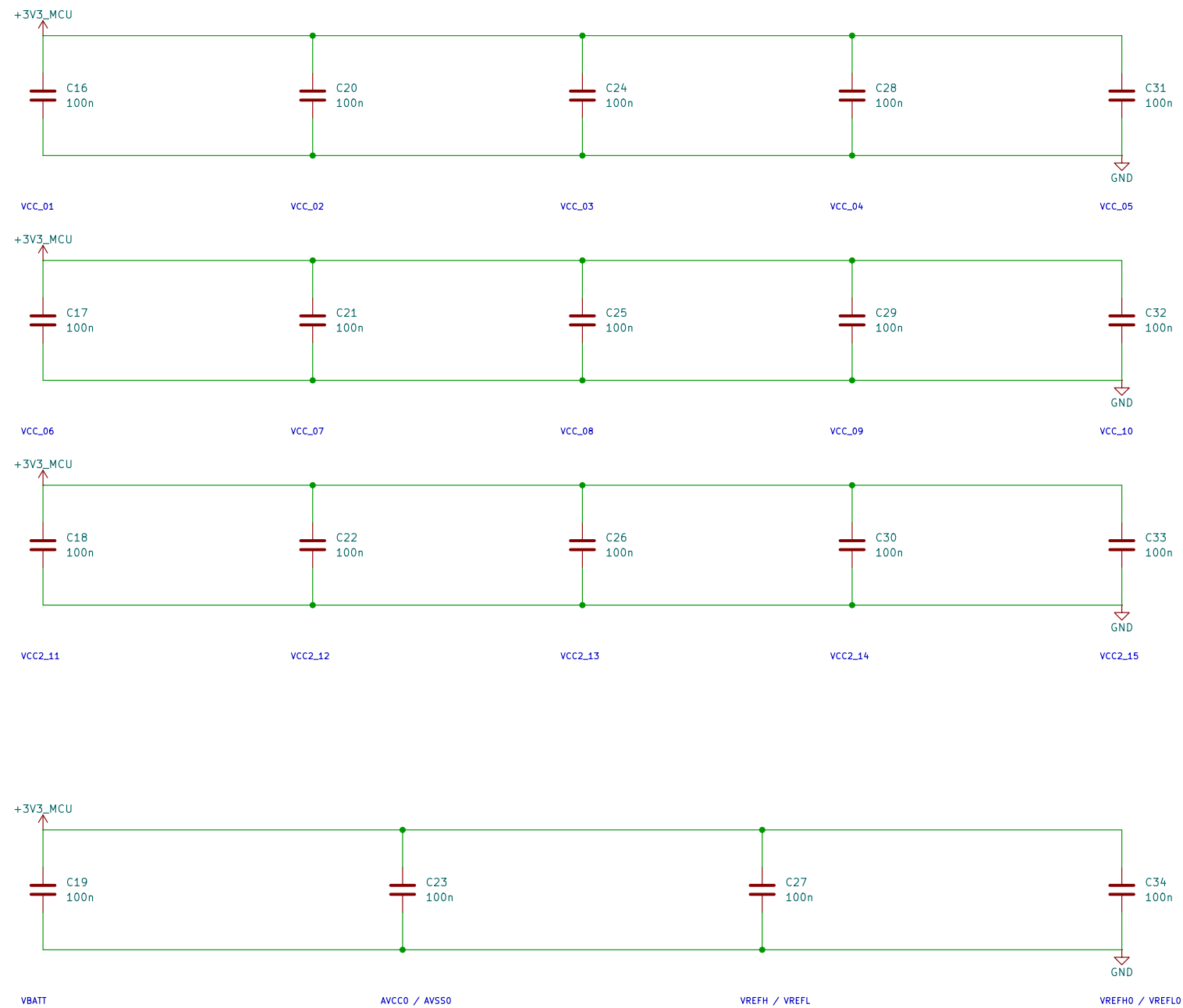
KiCad E.D.A. 10.0.5

Id: 2/8

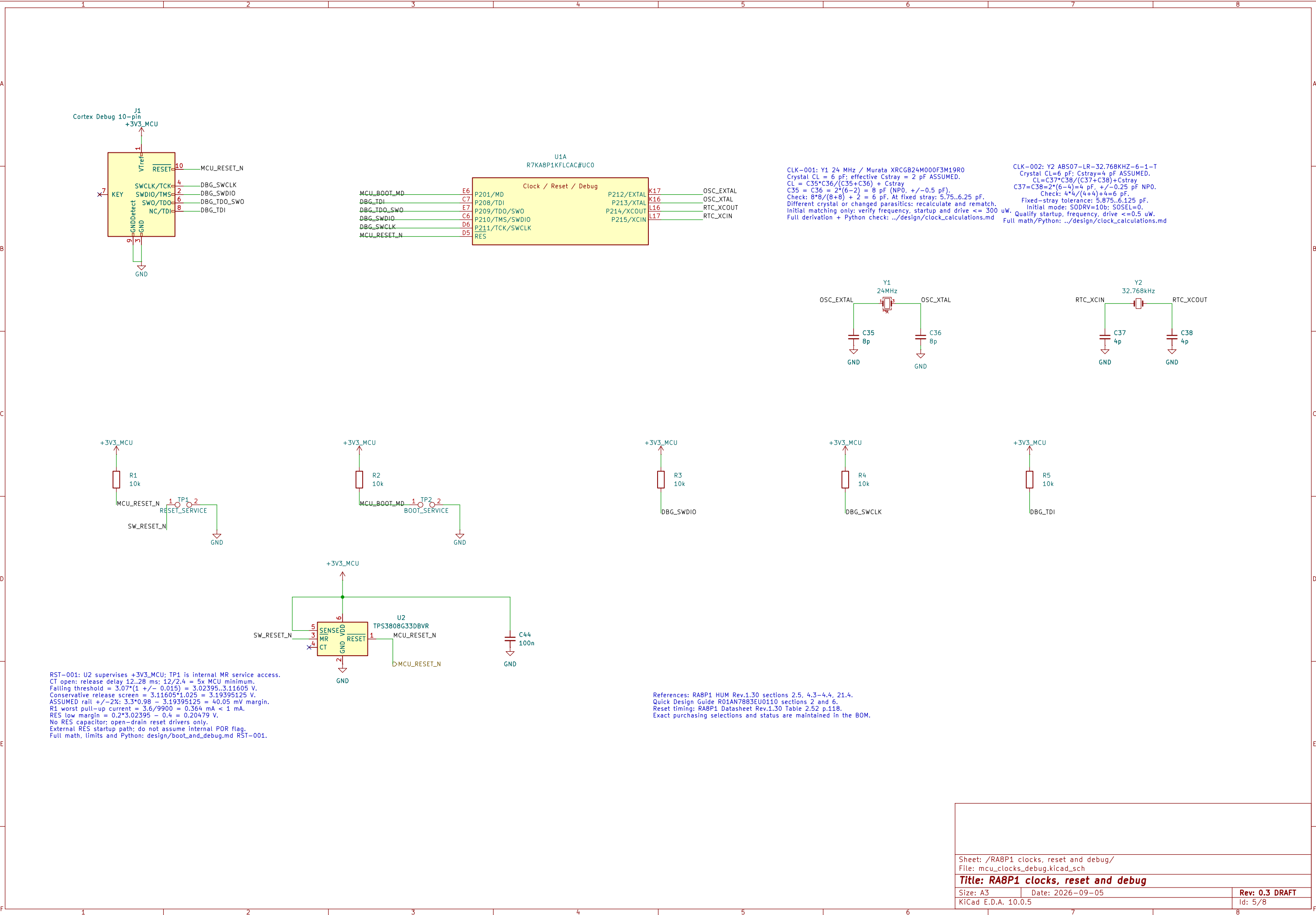


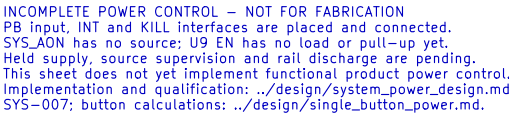


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIPI supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

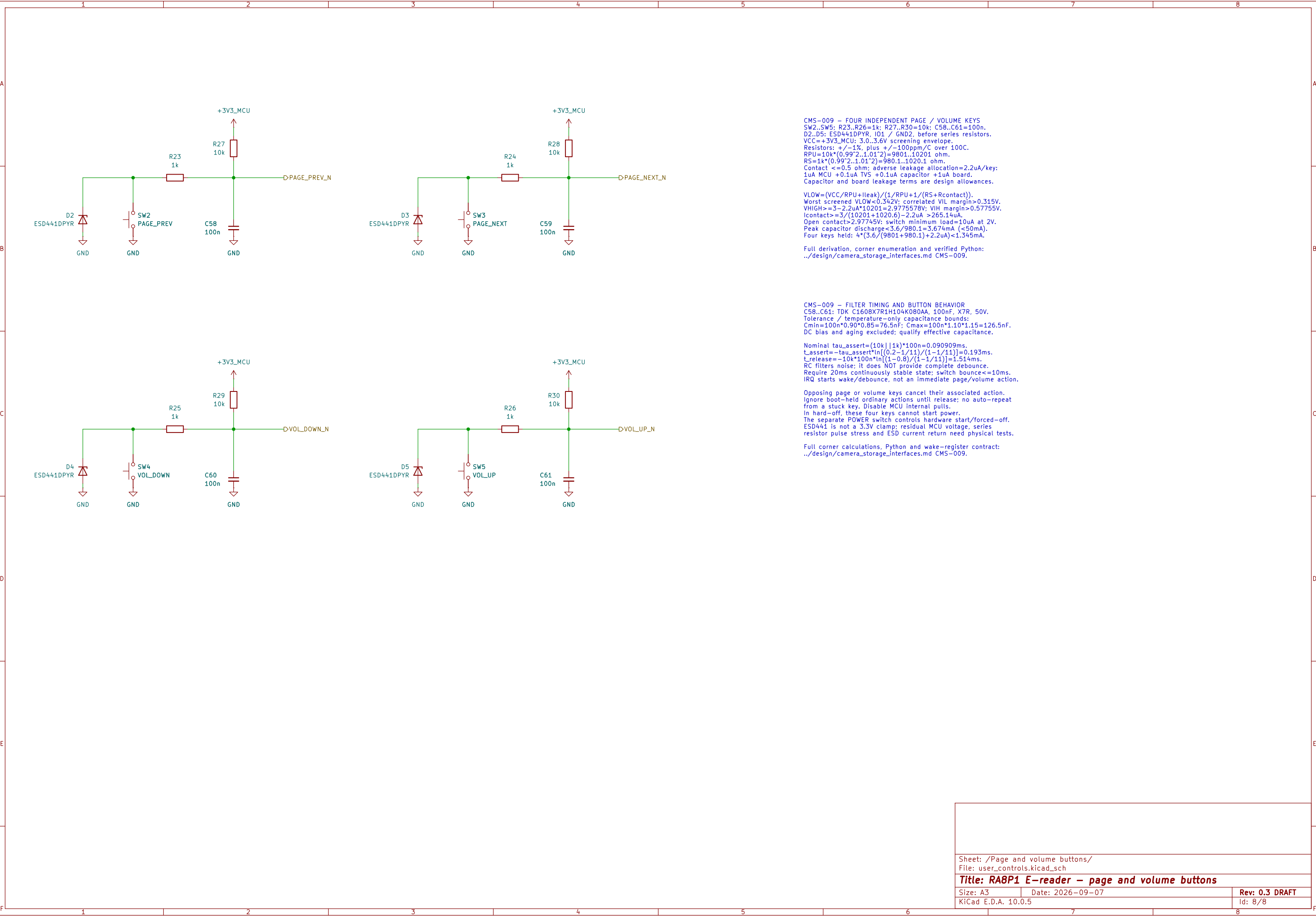




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BTN_004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF: TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
t_force=1e-6uF: t_Ladded[ms]=1/1.56e-4=6475.256ms (6.48s).
t_force_norm=64ms: t_force_norm=1ms + t_force_norm=64ms (6.48s).
C tolerance/temp: typ: 1*90%0.85=0.765uF.
1*1.10*1.15=1.265uF -> 4969.8174s using NOMINAL ic equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
Only in: internal test, not for production.
Release the initial ON press: after forced off, release then press again.
Full calculation + Python: ./design/single_button_power.m0 BTN_004/005.

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CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=980.1..1020.1 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact))$.
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(980.1+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.